Product Requirements Document: Telegram Smart Follow-Up Reminders

The "Action-Aware Assistant" for the Agentic Economy

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1. Executive Summary & Strategic Context

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Status: Portfolio Strategy / 2026 Roadmap

Date: February 17, 2026

1.1 Context

By 2026, the global messaging landscape has shifted from simple data exchange to **Agentic Autonomy**. AI agents now outnumber human users in digital workspaces, automating everything from inventory to B2B logistics.[1] However, this "bot-heavy" environment has created a **Human Commitment Gap**. In fast-moving chats, human users frequently make high-stakes promises—*"I'll send the UPI payment by 8 PM"* or *"Reviewing the contract now"*—which are instantly buried by the recency bias of active threads.[3]

1.2 The Vision

Upgrade Telegram from a conversation tool into a **Reliable Memory Layer**. By leveraging **Zero-Knowledge Edge AI**, we will enable users to capture, track, and execute chat-based commitments with zero friction. This feature positions Telegram not just as a WhatsApp competitor, but as a productivity competitor to Slack and Microsoft Teams in the high-growth SME markets of the Indian subcontinent.[5]

1.3 Problem Statement

- **User Pain:** Critical tasks are "lost in the scroll," leading to missed payments and damaged professional reliability.
- **Privacy Tension:** In 2026, server-side message scanning is legally untenable under the **India DPDP Act** and **EU AI Act**. Users demand intelligence without compromising the "Sealed Courier" ethos.
- **Market Gap:** Existing reminders are manual, command-based, or English-centric, ignoring the multilingual "Hinglish" nuances of 800M+ Indian users.

2. Competition Benchmarking

2.1 Competitive Landscape

In 2026, competitors are bifurcated between "dumb" privacy-first apps and "smart" cloud-heavy productivity tools.

2.2 Comparative Summary

Feature	Telegram (Proposed)	WhatsApp Business	Slack	Apple iMessage
Core Value	Action-Aware Memory	Mass Communication	Team Collaboration	Personal Continuity
Privacy Model	**Zero-Knowledge/Edge**	Cloud-Assisted AI	Cloud-Native	On-Device Static
Parsing Logic	Proactive Intent NLP	Manual Star/Pin	Command-based (/remind)	Keyword Detection
Indic Languages	**Bhashini-Integrated**	English-Centric	English-Centric	Limited Support
The "Hard" Moat	**On-device Multimodal**	Metadata Monetization	API Ecosystem	Hardware Lock-in

2.3 Recommendation

Telegram should double down on **Consent-Driven Intelligence**. Benchmarking shows that while Slack offers power, it creates "app fatigue." WhatsApp offers reach but lacks a dedicated task repository. Our recommendation is a **Centralized Action Center** powered by an on-device Edge-LLM. By integrating with **India's Bhashini API** for local dialects, we can capture the high-growth SME demographic that relies on "Hinglish" for business transactions .

3. Customer Experience (Core Workflow)

3.1 Personas & Segment Analysis

Persona	Motivation	High-Stakes Use Case
Rajesh (The Broker)	Scalability & Reliability	Receives property photos/addresses; needs to surface them during client calls.[3]
Neha (The Admin)	Community Efficiency	Tracks group-buy payments and volunteer deadlines for a 500-person thread.[3]
SME Owner (Kirana)	Cash-Flow Management	Tracks vendor credit promises and delivery dates in regional dialects .

3.2 The Interaction Model (Wireframe Descriptions)

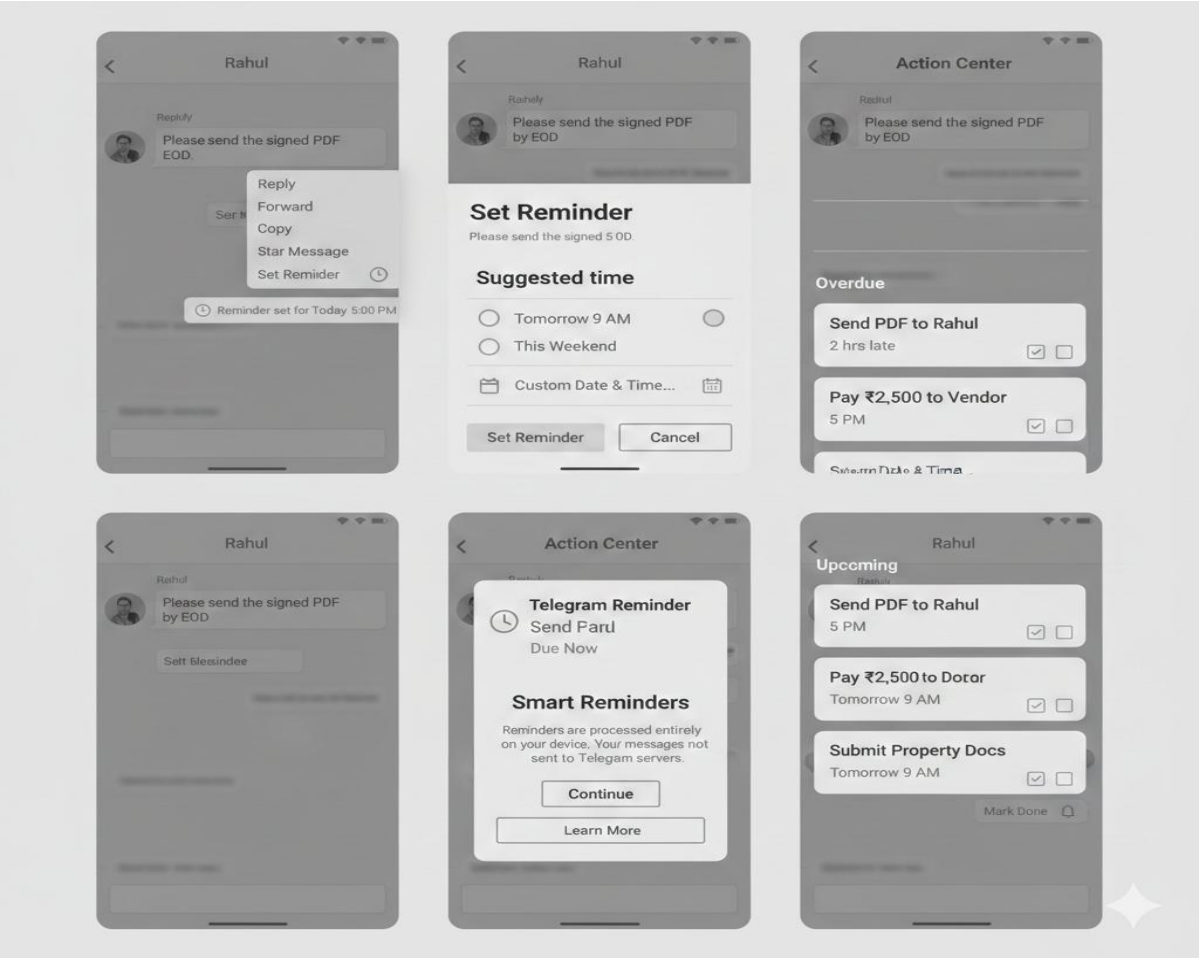
Feature Component	CX Flow	Technical Acceptance Criteria (TAC)
Smart Capture	Long-press a message -> Select "Set Reminder."	Intent Parser must extract Action + Date + Target with >95% precision .
The Preview Card	A bottom-sheet pops up with AI-parsed fields (e.g., "Send PDF" @ 5:00 PM).	Latency from long-press to card display must be <150ms on mid-range devices .
Contextual Nudge	A toast notification surfaces in the chat: "You told Arjun 'EOD'—4 hours left."	Nudges must include a "Deep-Link" back to the specific message ID in TDLib.
Multilingual Switch	User types in Hinglish ("Kal pakka payment karunga").	System must leverage Bhashini API for PII-masked translation of regional intent .

4. Detailed Feature Specifications (Action Center)

4.1 The "Action Center" UI

Instead of burying reminders in a "Saved Messages" chat, the feature introduces a dedicated **Action Center Tab**.

- **View 1: Proximity Sort.** Items sorted by "Minutes to Deadline."
- **View 2: Chat-Origin Grouping.** Clusters commitments by the person they were made to.
- **View 3: Actionable Widgets.** A "One-Tap Pay" button for UPI-related reminders, or "Attach File" for document-related tasks



4.2 Edge-LLM Orchestration

The parsing engine will utilize a **distilled Small Language Model (SLM)** (e.g., a variant of Llama-3-8B or Qwen-0.5B optimized for Indic languages).

- **Constraint:** The model must reside in the local app binary to ensure **Zero-Knowledge**—Telegram servers never receive the plaintext of the message being "reminded".

4.3 Multimodal Context

If a user long-presses a **Voice Note** or **Image** (e.g., a photo of a check), the system triggers:

1. **On-Device STT/OCR:** Converts audio or image text into a temporary string.
2. **Intent Parsing:** Extracts the commitment from the transcribed text.

5. Technical Considerations & Regulatory Compliance

5.1 Privacy-First Architecture (The "Principal" Moat)

In 2026, trust is the primary competitive advantage. The architecture is built on three pillars:

1. **On-Device Processing:** Using **TDLib (Telegram Database Library)** to store reminder metadata locally. Cloud sync only happens for **Encrypted Blobs** where Telegram holds no keys .
2. **Purpose Limitation (DPDP Act):** Data processed for the reminder is *transient*. Once the reminder is set, the parsed text is purged from the AI's context window .
3. **Article 50 Transparency (EU AI Act):** Every reminder card will carry a "System-Generated" label to ensure users know they are interacting with an AI suggestion .

5.2 Success Metrics (The KPI Framework)

Metric	Business Definition	Target (2026)
ACR (Action Completion Rate)	% of set reminders that result in a task-completion action (e.g., file sent, UPI link clicked).	**> 72%**
Multilingual Precision	Accuracy of parsing Hinglish/Tamil-English intent vs. English baseline.	**> 88%**
Friction Index	Time from "Long-press" to "Reminder Confirmed" by user.	**< 2.1 seconds**
Privacy Opt-in Rate	% of users who enable the Edge-AI parser after reading the DPDP consent notice.	**> 90%**

6. Roadmap & GTM Strategy

6.1 Prioritized Phasing

Priority	Feature	Impact	Effort	Timeline
P0	**Edge-Parser (English)**	Critical for core utility.	High	Week 1-6
P0	**Action Center (Tab 4)**	Essential for task management.	Medium	Week 2-8
P1	**Bhashini Hinglish Sync**	Massive growth lever for India.	High	Week 10-16
P1	**Deep-Link Context Nudges**	Primary retention driver.	Medium	Week 8-12
P2	**Voice/Image (OCR/STT)**	Expansion into multimodal.	High	Week 18+

6.2 Prioritization Reasoning

We prioritize the **English Edge-Parser** and **Action Center** first to prove the "Zero-Knowledge" technical feasibility. **Hinglish support** is a P1 rather than P0 due to the complexity of local hardware testing across the Indian subcontinent's fragmented Android ecosystem.

7. Appendix & Research Notes

7.1 Technical References

- **TDLib Integration:** Reminders will be stored as EncryptedLocalObject within the TDLib structure to ensure cross-platform metadata sync without server-side decryption.
- **Model Compression:** Utilizing **4-bit Quantization** to ensure the Indic-SLM fits within <500MB of RAM for mid-range Indian devices.
- **Zero-Knowledge Proofs (ZKP):** Considering ZKPs for future "Shared Reminders" to verify a task is done without revealing chat contents to the group admin.

7.2 Open Issues & Risks

- **Model Drift:** How does the on-device model handle evolving Indian slang?
  - *Mitigation:* Monthly "Differential Privacy" model weight updates via **Federated Learning** .
- **Battery Impact:** Continuous background parsing for "Auto-Suggestions."
  - *Decision:* **Features Out** - We will not support proactive background scanning in Phase 1 to

preserve battery and privacy trust.[8]

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